Decrease in anti-Proteus mirabilis but not anti-Escherichia coli antibody levels in rheumatoid arthritis patients treated with fasting and a one year vegetarian diet.

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OBJECTIVE: To measure Proteus mirabilis and Escherichia coli antibody levels in patients with rheumatoid arthritis (RA) during treatment by vegetarian diet. METHODS: Sera were collected from 53 RA patients who took part in a controlled clinical trial of fasting and a one year vegetarian diet. P mirabilis and E coli antibody levels were measured by an indirect immunofluorescence technique and an enzyme immunoassay, respectively. RESULTS: The patients on the vegetarian diet had a significant reduction in the mean anti-proteus titres at all time points during the study, compared with baseline values (all $p < 0.05$). No significant change in titre was observed in patients who followed an omnivorous diet. The decrease in anti-proteus titre was greater in the patients who responded well to the vegetarian diet compared with diet non-responders and omnivores. The total IgG concentration and levels of antibody against E coli, however, were almost unchanged in all patient groups during the trial. The decrease from baseline in proteus antibody levels correlated significantly ($p < 0.001$) with the decrease in a modified Stoke disease activity index. CONCLUSION: The decrease in P mirabilis antibody levels in the diet responders and the correlation between the decrease in proteus antibody level and decrease in disease activity supports the suggestion of an aetiopathogenetic role for P mirabilis in RA.